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[22AECME51]

FIFTH SEMESTER B. TECH EXAMINATION, DECEMBER-2024

CONDITION MONITORING AND MAINTENANCE MANAGEMENT

Time: 3 Hours **Maximum Marks: 100**

Instructions: i. Missing data may be suitably assumed.

ANSWER ALL QUESTIONS

		PART-A			**5 x2=10**
1.	**a.**	Write the functions of maintenance	L1	CO1	**02**
	b.	Define Run-out Maintenance.	L1	CO2	**02**
	c.	List the goals of TPM.	L1	CO3	**02**
	d.	Define fishbone diagram.	L1	CO4	**02**
	e.	Define digital signal processing.	L1	CO5	**02**
		PART-B			**6 X 5=30**
2.	**a.**	Explain spare parts management system.	L2	CO1	**05**
		OR			
	b.	Discuss on maintenance policy goals.	L2	CO1	**05**
3.	**a.**	Discuss probability theory a with examples in maintenance management.	L2	CO2	**05**
		OR			
	b.	Write short on RCM.	L2	CO2	**05**
4.	**a.**	Explain Pareto analysis.	L2	CO3	**05**
		OR			
	b.	Explain key performance indicators of TPM.	L2	CO3	**05**
5.	**a.**	Prepare the preventative maintenance checklist for two wheeler.	L3	CO4	**05**
		OR			
	b.	Explain subjective condition monitoring and objective condition monitoring.	L2	CO4	**05**
6.	**a.**	Discuss fault diagnosis for rotational machines.	L3	CO5	**05**
		OR			
	b.	Prepare a FMEA chart for mobile phones.	L3	CO5	**05**
7.	**a.**	Discuss on the development of maintenance system.	L2	CO1	**05**
		OR			

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	b.	Explain MTTR and MTBF.	L2	CO2	05
		PART-C	**6 X 10=60**		
8.	a.	Discuss predictive maintenance with example.	L2	CO1	10
		OR			
	b.	List different types of maintenance strategies and explain any one with examples	L2	CO1	10
9.	a.	Explain bathtub curve for electric blub.	L2	CO2	10
		OR			
	b.	Write the procedure for conducting RCM analysis	L2	CO2	10
10.	a.	Interpret 5's for the food package industries.	L3	CO3	10
		OR			
	b.	Sketch and explain fishbone diagram for cellular phone.	L3	CO3	10
11.	a.	Explain the condition monitoring process for food package industries.	L3	CO3	10
		OR			
	b.	Explain the levels of condition monitoring with example.	L3	CO3	10
12.	a.	Write different methods of Condition monitoring techniques and any one with neat sketch	L3	CO5	10
		OR			
	b.	Explain digital signal processing and computational techniques in maintenance system.	L3	CO5	10
13.	a.	Explain 8 pillars of TPM.	L5	CO6	10
		OR			
	b.	Explain **i.** Failure probability **ii.** Redundancy **iii.** Hazard rate function **iv.** MTTR **v.** MTBF	L5	CO6	10